

Bridging the Gap: A Review of AI Weather Forecasting Limitations and Hybrid Solutions for Extreme Events

The integration of deep learning into meteorology has initiated a "second revolution," offering computationally efficient alternatives to traditional Numerical Weather Prediction (NWP) models. However, a critical dichotomy exists: while AI models excel in global skill metrics, they frequently fail to capture the intensity and non-linear dynamics of high-impact extreme events. This review synthesizes current literature highlighting that symmetric loss functions and reliance on historical reanalysis data cause deterministic AI models to regress toward climatology, systematically underestimating record-breaking heat, wind, and precipitation [5] [102]. Furthermore, these models struggle with "gray swan" events, lacking the physical inductive biases necessary for extrapolation beyond their training distributions [25]. To address these deficiencies, the field is shifting toward hybrid frameworks that integrate physics-informed constraints, asymmetric loss functions, and generative architectures like diffusion models to restore fine-scale details and improve probabilistic consistency [17] [163]. Despite advancements in specialized benchmarks such as the Extreme Weather Bench, significant gaps remain in standardizing verification for operational reliability. This review argues that future progress depends on moving beyond pure data-driven emulation to robust, hybrid systems that leverage AI efficiency while ensuring the physical fidelity required for trustworthy early warning systems in a changing climate.

Introduction: The Paradigm Shift and Verification Challenges in AI Weather Forecasting

The integration of deep learning into meteorology has initiated a transformative "second revolution" in weather forecasting, offering computationally efficient alternatives to traditional Numerical Weather Prediction (NWP) models while simultaneously introducing significant challenges regarding reliability for high-impact extremes. Historically, operational forecasting has relied on solving complex partial differential equations governing atmospheric dynamics, a process that is computationally expensive and often limited by resolution constraints [222]. In contrast, data-driven Deep Learning Weather Prediction (DLWP) models have emerged as powerful tools capable of learning spatio-temporal dynamics directly from reanalysis data, achieving forecast skills comparable to or exceeding state-of-the-art NWP systems at a fraction of the computational cost [38]. Models such as Pangu-Weather, GraphCast, and FengWu have demonstrated superior performance in medium-range global forecasting, effectively emulating the atmospheric state with high accuracy across various variables including geopotential, temperature, and wind speed [10]. However, this rapid advancement has exposed a critical dichotomy: while AI models excel in general skill metrics like Root Mean Square Error (RMSE) and Anomaly Correlation Coefficient (ACC), they frequently struggle to capture the non-linear, rare, and intense characteristics of extreme weather events [16].

A primary concern in the current literature is the tendency of deep learning models to produce "smooth" forecasts that underestimate the intensity of extremes, a phenomenon often attributed to the widespread use of symmetric loss functions such as Mean Squared Error (MSE) during training [102]. The optimization of MSE inherently favors the prediction of conditional means, leading to a regression toward climatology and a failure to resolve the long-tailed distributions characteristic of precipitation and temperature extremes [46]. Consequently, while AI models may accurately predict the synoptic-scale structure of a storm, they often fail to resolve the fine-scale features necessary for issuing precise warnings, such as peak wind gusts or localized heavy precipitation [60]. This limitation is particularly pronounced in record-breaking events, where numerical models like the ECMWF High-Resolution Forecast (HRES) consistently outperform AI counterparts by better extrapolating beyond the training domain [5]. The inability of current AI models to reliably forecast unprecedented extremes raises questions about their physical consistency and their capacity to maintain atmospheric balances during high-impact scenarios [123].

To address these deficiencies, the research community is increasingly focusing on specialized architectures, loss functions, and hybrid approaches designed explicitly for extreme event prediction. Recent studies highlight the efficacy of asymmetric loss functions and custom training objectives that penalize errors in extreme quantiles more heavily, thereby reducing the smoothing effect and improving the representation of tail events [6]. Furthermore, generative modeling techniques, including Generative Adversarial Networks (GANs) and diffusion models, are being employed to restore fine-scale details and generate realistic ensemble members that capture the stochastic nature of extremes [17]. For instance, the introduction of denoising diffusion probabilistic models has shown promise in enhancing the resolution of surface forecasts, effectively correcting the underestimation of extreme precipitation and wind speeds observed in deterministic AI models [17]. Additionally, physics-informed machine learning approaches are gaining traction, integrating physical constraints and domain knowledge to ensure that predictions remain dynamically consistent even in rare event regimes [141].

Despite these methodological advancements, rigorous verification remains a significant bottleneck. Traditional global-scale evaluations often mask local failures in extreme event prediction, necessitating the development of impact-centric benchmark suites and case-study-driven assessments [2]. Initiatives such as the Extreme Weather Bench (EWB) and datasets like HR-EXTREME are crucial for standardizing the evaluation of AI models against high-impact hazards, revealing that errors in extreme cases are significantly larger than overall forecast errors [24]. Moreover, the scarcity of historical data for rare events poses a fundamental challenge, often requiring the use of long climate model simulations or transfer learning strategies to adequately train models on extreme regimes [86]. The interplay between data scarcity, model architecture, and physical consistency defines the current frontier of AI weather forecasting, where the goal is not merely to match NWP performance on average metrics but to surpass it in the prediction of the most societally relevant and dangerous weather phenomena [124]. As the field evolves, the focus is shifting from purely data-driven emulation to hybrid systems that leverage the speed of AI and the physical robustness of NWP, ensuring that the "second revolution" delivers reliable warnings for a changing climate [34].

The Need for Standardized Benchmarking and Datasets

The rapid evolution of Artificial Intelligence (AI) in weather forecasting has precipitated a paradigm shift, yet the field currently suffers from significant fragmentation in model evaluation. While AI models are increasingly surpassing traditional Numerical Weather Prediction (NWP) systems in global metrics, their reliability in forecasting high-impact extreme events remains uncertain and inconsistently verified. A primary driver of this uncertainty is the reliance on evaluation frameworks that prioritize general accuracy over extreme event fidelity, often utilizing global-scale metrics or hand-picked case studies that fail to capture the full spectrum of atmospheric hazards [2]. To address these deficiencies, the community is moving toward the development of standardized, community-driven benchmarks and high-resolution datasets specifically designed to target extreme phenomena, thereby enabling rigorous, comparable, and impact-centric verification.

A critical limitation of existing benchmarks is their dependence on reanalysis products, such as ERA5, which are generated through delayed data assimilation and do not reflect the constraints of real-time operational forecasting. This creates a systematic mismatch between benchmark performance and real-world utility, particularly for extreme events where latency and data availability are crucial [72]. Furthermore, standard metrics like Mean Squared Error (MSE) inherently bias models toward average conditions, leading to the underestimation of extreme values and a smoothing of forecast fields as lead times increase [102]. Consequently, models that perform well on global averages may fail catastrophically when tasked with predicting record-shattering heatwaves or intense precipitation events, as evidenced by studies showing that AI models often underestimate the frequency and intensity of record-breaking extremes compared to high-resolution NWP systems [5].

To rectify these issues, new benchmark suites are being introduced to facilitate true cross-model comparisons on high-impact phenomena. The Extreme Weather Bench (EWB) represents a significant step in this direction, offering a standard set of case studies spanning multiple spatial and temporal scales, alongside observational data and impact-based metrics [2]. Similarly, RealBench emphasizes realistic evaluation under operational conditions by integrating low-latency operational analysis and large-scale in-situ observations, revealing substantial discrepancies between reanalysis-based metrics and real-world performance [72]. These initiatives are complemented by specialized datasets like HR-EXTREME, which provides high-resolution extreme weather cases derived from NOAA's 3-km HRRR data, addressing the scarcity of curated datasets for such events and highlighting that errors in extreme cases are significantly larger than overall forecast errors [24].

The development of these resources is driven by findings that AI models struggle with extrapolation beyond their training domains, particularly for "gray swan" events that are absent from historical training data. For instance, research indicates that AI models trained without major tropical cyclones cannot accurately forecast Category 5 storms, demonstrating an inability to extrapolate from weaker events [25]. This limitation underscores the necessity for benchmarks that specifically test out-of-distribution generalization and compound impact metrics, which are often overlooked in standard evaluations [10]. Moreover, the lack of standardized evaluation has led to inconsistent conclusions regarding model superiority; while some studies suggest AI models match NWP performance on specific heatwaves, others show they underperform when errors are aggregated over space and time or when assessing compound winter storms [10].

Beyond static datasets, there is a growing recognition of the need for dynamic and multimodal benchmarking frameworks. The CLLMate benchmark addresses the gap in translating numerical variables into actionable textual narratives, while MPBench provides a large-scale temporal multimodal dataset for severe weather event prediction, aligning raw meteorological data with text captions [199][250]. Additionally, frameworks like EWE introduce benchmarks for automated diagnostic reasoning, moving beyond mere prediction to the interpretation of physical mechanisms driving extreme events [27]. The integration of these diverse benchmarks is essential for developing robust loss functions and training strategies, such as asymmetric optimization and extreme value loss regularization, which have been shown to improve hit rates for low-probability events [102][65].

Ultimately, the transition from general accuracy metrics to specialized, extreme-focused benchmarking is vital for building trust in AI weather models. Without standardized evaluation protocols that prioritize high-impact scenarios, the deployment of these models in early warning systems and disaster management remains precarious. The consensus

across recent literature is that future progress depends on the continuous evolution of these community-driven systems to include additional phenomena, test cases, and metrics that reflect the complex, non-linear nature of extreme weather in a changing climate [2][124]. By establishing rigorous standards, the research community can ensure that advancements in AI forecasting translate into tangible improvements in societal resilience against extreme weather hazards.

Limitations of Current Evaluation Metrics

The rapid advancement of Artificial Intelligence (AI) in weather forecasting has precipitated a paradigm shift, yet traditional evaluation metrics often fail to capture the nuanced societal impacts of compound events or the specific dynamics of rare occurrences. This discrepancy leads to a potential overestimation of AI model readiness for critical disaster management applications. While AI models such as Pangu-Weather and GraphCast have demonstrated competitive skill against Numerical Weather Prediction (NWP) systems on global benchmarks, these assessments frequently rely on aggregate statistics like Root Mean Square Error (RMSE) over reanalysis datasets, which can mask significant deficiencies in predicting high-impact extremes [74; 5]. A primary limitation is the tendency of data-driven models to regress toward climatology, resulting in the underestimation of the intensity and frequency of record-breaking events. For instance, studies indicate that state-of-the-art AI models consistently underpredict hot records and overestimate cold records, with errors growing as the magnitude of the record exceedance increases, thereby compromising their utility for early warning systems in a rapidly warming climate [5; Forecasting the Future with Yesterday's Climate: Temperature Bias in AI Weather and Climate Models].

Furthermore, standard verification frameworks often overlook the physical consistency and structural realism required for impact-based decision-making. Although AI models may accurately capture synoptic-scale structures, they frequently fail to resolve the fine-scale details necessary for issuing precise weather warnings. In the case of Storm Ciara, while AI models correctly identified the large-scale dynamical drivers, they underestimated peak wind amplitudes and failed to capture sharp frontal gradients, which are critical for assessing local damage potential [60]. This smoothing effect is exacerbated at longer lead times; research on the Aurora model reveals that while large-scale circulation patterns remain skillful at subseasonal scales, threshold-based extreme intensity collapses beyond seven to ten days as fields revert to climatological means [13]. Consequently, metrics that prioritize mean state accuracy over tail behavior provide a misleading picture of operational reliability.

The challenge is further compounded by the inability of current metrics to adequately assess model performance on "gray swan" events—plausible but unobserved extremes that lie outside the training distribution. Traditional evaluation relies heavily on historical data, yet AI models struggle to extrapolate from weaker events in their training sets to stronger, unseen extremes. Experiments removing Category 3-5 tropical cyclones from training data demonstrate that models cannot accurately forecast Category 5 storms, indicating a fundamental lack of extrapolation capability that standard skill scores do not sufficiently penalize [25; Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall]. Moreover, the reliance on reanalysis products like ERA5 for benchmarking introduces a systematic mismatch with real-world operational conditions, as these products incorporate delayed data assimilation that is unavailable in real-time forecasting scenarios [72]. This gap suggests that high scores on retrospective benchmarks may not translate to effective real-time disaster response.

Another critical oversight in current evaluation methodologies is the neglect of compound events and multivariate dependencies. High-impact weather often manifests through the confluence of multiple hazards, yet many AI models exhibit weaker error correlations between atmospheric variables, suggesting they may overlook physical constraints essential for modeling compound risks [112]. Validating Deep-Learning Weather Forecast Models on Recent High-Impact Extreme Events highlights that while ML models may achieve local accuracy comparable to NWP on specific variables, they often underperform when errors are aggregated over space and time or when assessing compound winter storms. Additionally, the lack of robust uncertainty quantification in deterministic AI models poses a significant risk; without reliable probabilistic outputs, decision-makers cannot adequately gauge the confidence levels associated with extreme event forecasts, a gap that traditional deterministic metrics fail to address [9; 78].

To address these shortcomings, the community is moving toward impact-centric and threshold-weighted verification approaches. New frameworks propose weighting scoring rules to emphasize high-impact outcomes and utilizing event-based metrics that better reflect user needs, such as the Extreme Weather Bench (EWB) which focuses on high-impact hazards rather than global averages [2; 139; 130]. These approaches aim to bridge the gap between statistical skill and societal relevance, ensuring that model evaluations account for the specific dynamics of rare occurrences and the cascading effects of compound disasters. However, until these specialized metrics become standard, the prevailing reliance on traditional aggregate scores risks overstating the readiness of AI systems for the high-stakes domain of disaster management.

Comparative Performance: AI vs. Numerical Models in Extreme Scenarios

The dichotomy between the performance of Artificial Intelligence (AI) weather models and traditional Numerical Weather Prediction (NWP) systems becomes most pronounced when forecasting unprecedented or structurally complex extreme events. While AI models such as Pangu-Weather and GraphCast have demonstrated superior accuracy over NWP benchmarks for standard atmospheric trajectories and large-scale patterns under ordinary conditions, they consistently exhibit limitations when confronted with record-breaking extremes and sudden structural changes. A primary concern identified in recent literature is the tendency of data-driven models to regress toward climatology, leading to a systematic underestimation of event intensity. For instance, studies comparing state-of-the-art AI models against the European Centre for Medium-Range Weather Forecasts (ECMWF) High RESolution forecast (HRES) reveal that numerical models consistently outperform AI counterparts in predicting record-breaking heat, cold, and wind events. These AI models not only underestimate the frequency and intensity of such records but also display growing errors as the magnitude of the record exceedance increases, highlighting a fundamental struggle with extrapolation beyond their training domains [5].

This limitation is further evidenced in the behavior of AI models during subseasonal forecasting horizons. While models like Aurora demonstrate strong short-range skills across various event types, including tropical cyclones and atmospheric rivers, a consistent failure mode emerges at lead times beyond seven to ten days. In these extended ranges, while large-scale circulation patterns remain moderately skillful, the threshold-based intensity of extreme events collapses as the model fields regress toward climatological means. This divergence suggests that while AI models retain synoptic-scale dynamical structure, they lose the surface-impact amplitude necessary for accurate high-impact warnings [13]. Similarly, evaluations of heat waves indicate that AI models often exhibit cold biases prior to and during extreme heat events, although some models like GraphCast show improved skill compared to traditional ensemble systems in specific regional contexts [18]. The issue of smoothing is intrinsic to the training objectives of many deterministic AI models; the reliance on symmetric loss functions, such as Mean Squared Error, inherently biases predictions toward the mean, resulting in "blurry" forecasts that fail to capture the sharp gradients and peak amplitudes characteristic of severe storms [102].

Despite these general trends, the performance landscape is nuanced, with AI models showing competitive or superior capabilities in specific structural scenarios. For example, in the case of the unprecedented 2024 Dubai rainfall, GraphCast successfully forecasted the event eight days in advance, not through extrapolation from weaker training events, but via "translocation," leveraging dynamically similar events from other regions within its global receptive field. Conversely, other models like FuXi underestimated the rainfall intensity, illustrating the variability in how different architectures handle "gray swan" events [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall]. Furthermore, AI models have shown remarkable proficiency in predicting the tracks of tropical cyclones and the mean sea level pressure of storm series, occasionally outperforming NWP models on weekly timescales. However, this success in track prediction does not always translate to intensity forecasting; AI models frequently underestimate peak wind speeds and fail to resolve fine-scale structures such as warm core seclusion or sharp frontal gradients, which are critical for issuing precise impact warnings [112] [60].

The inability of AI models to extrapolate from weaker to stronger events is particularly evident in tropical cyclone intensity prediction. Research indicates that models trained without exposure to major hurricanes (Category 3-5) fail to accurately forecast Category 5 "gray swan" cyclones, confirming that current learning strategies cannot reliably extrapolate intensity from weaker analogs. This limitation persists even when models are trained on global data, suggesting that regional information is implicitly encoded but insufficient for out-of-distribution generalization in extreme regimes [25] [243]. Consequently, while AI models excel in capturing ordinary TC trajectories, they lag behind NWP systems in predicting rare, sudden-turning tracks where fine structural interactions between the storm core and steering flows are paramount [1].

To mitigate these deficiencies, recent research advocates for hybrid approaches and specialized architectural modifications. Techniques such as incorporating terrain-following coordinates and conservation schemes into AI

models have shown promise in reducing drizzle bias and improving the estimation of extreme precipitation spectra [INVESTIGATING THE CONTRIBUTION OF TERRAIN-FOLLOWING COORDINATES AND CONSERVATION SCHEMES IN AI-DRIVEN PRECIPITATION FORECASTS]. Additionally, the integration of diffusion models and generative frameworks, such as GenCast and FuXi-Extreme, aims to restore fine-scale details and generate realistic ensemble spreads that deterministic models miss, thereby better capturing the tails of probability distributions for extreme events [163] [17]. Nevertheless, the consensus remains that while AI models offer transformative speed and efficiency, their standalone reliability for high-stakes disaster management involving unprecedented extremes is currently limited by their dependence on historical data distributions and their tendency to smooth out the very anomalies that define catastrophic weather [10] [24].

Tropical Cyclone Track and Intensity Prediction

The comparative performance of Artificial Intelligence Weather Prediction (AIWP) models versus traditional Numerical Weather Prediction (NWP) systems reveals a nuanced landscape where data-driven approaches excel in standard scenarios but encounter significant limitations during complex, extreme dynamical transitions. Recent evaluations indicate that AI models generally outperform NWP in predicting ordinary tropical cyclone (TC) tracks and looping trajectories, yet they frequently fail to accurately capture sudden-turning paths driven by complex interactions with weak steering flows. This dichotomy is starkly illustrated in analyses of Northwest Pacific typhoons from 2020 to 2024, where models such as Pangu-Weather demonstrated superior accuracy for ordinary TCs and looping cases compared to the ECMWF-IFS, reducing mean distance errors by 13.5% and 32.9% respectively [1]. However, this advantage evaporates when confronting sudden-turning events; for these rare trajectories, Pangu-Weather exhibited track forecast errors averaging 9.4% greater than those of ECMWF-IFS, particularly struggling with storms like Typhoon Khanun which executed sharp turns under weak subtropical high conditions [1].

The underlying mechanism for this discrepancy appears rooted in the fundamental differences between how AI and NWP models process atmospheric dynamics. AIWP models, trained on historical reanalysis data, are highly adept at recognizing and extrapolating large-scale circulation patterns, such as a strong Western Pacific Subtropical High (WPSH), which typically governs ordinary TC motion [1]. Conversely, sudden turns often occur when the steering flow is weak, forcing the TC trajectory to be determined by the intricate interaction between the storm's internal structure and the immediate environment. In these regimes, the failure of AI models to realistically simulate fine-scale structural characteristics and reproduce the physics of the TC core region leads to degraded forecast skill [1]. This limitation is further corroborated by studies on physical consistency, which show that regional deep learning models can disrupt key atmospheric balances and generate unrealistic spatial gradients during severe extratropical cyclones, suggesting a lack of adherence to the governing fluid equations that NWP models strictly enforce [123].

Despite these challenges in specific track morphologies, AI models have demonstrated remarkable utility in other aspects of TC prediction, particularly regarding computational efficiency and ensemble generation. The rapid inference speed of AI models facilitates the creation of large ensembles, allowing for robust probabilistic forecasting that was previously computationally prohibitive. For instance, perturbation-based methods using the Pangu model can generate thousands of ensemble scenarios rapidly, aligning closely with ECMWF ensemble predictions up to seven days prior to landfall [74]. Furthermore, AI-based Conditional Nonlinear Optimal Perturbation (CNOP) frameworks have successfully identified rapidly growing error patterns with physical explainability, offering a low-cost alternative to traditional NWP optimization for predictability analysis [39]. Sensitivity analyses using Finite-Time Lyapunov Exponents (FTLE) within AI frameworks have also highlighted that trajectory sensitivity is not uniformly distributed but concentrates at specific "recurvature gates," where small perturbations in the steering environment can lead to catastrophic amplification of track deviations [244].

Regarding intensity prediction, the performance gap between AI and NWP models has historically been wider, with early AIWP models exhibiting a substantial low bias and often reducing TC intensity within the first 24 hours of the forecast [243]. Global AI models trained on coarse-resolution reanalysis data often smooth out the extrema associated with TCs, leading to underestimations of peak winds and central pressure [3]. However, recent advancements in hybrid modeling and post-processing are beginning to bridge this gap. Hybrid systems that integrate ML large-scale forecasts with physics-based regional models, such as the FuXi-SHTM framework, have shown clear improvements, reducing intensity forecast errors by nearly 60% compared to standalone ML models [76]. Similarly, specialized transformer-based models like TIFNet, which integrate high-resolution global forecasts with historical evolution mechanisms, have achieved robust improvements in rapid intensification regimes, reducing errors by 29-43% relative to operational baselines [276].

Nevertheless, the issue of extrapolation remains a critical vulnerability for purely data-driven approaches, particularly concerning "gray swan" events—extremes that are absent from the training dataset. Studies indicate that AI models trained without Category 3-5 TCs struggle to forecast Category 5 events, suggesting an inability to extrapolate from weaker storms to unseen extremes [25]. This limitation extends to record-breaking extremes, where numerical models like the ECMWF HRES consistently outperform state-of-the-art AI models in forecasting unprecedented heat, cold, and wind events [5]. While diffusion models and generative approaches offer promising avenues for restoring fine-scale

details and generating realistic extreme samples, the consensus indicates that current AI formulations require improved bias corrections, higher-resolution training data, or hybridization with physical constraints to reliably predict the full spectrum of TC behaviors, especially those driven by non-linear, weak-flow interactions [30].

Heatwave and Temperature Extremes

The integration of Artificial Intelligence (AI) into weather forecasting has precipitated a paradigm shift, yet the reliable prediction of heatwaves and temperature extremes remains a complex challenge characterized by systematic biases and extrapolation limitations. A predominant issue identified across multiple state-of-the-art models is a consistent cold bias, particularly evident in the days leading up to extreme heat events. Research indicates that AI-based models, including FourCastNet V2 Small, Pangu Weather, and the Ai2 Climate Emulator version 2, frequently produce cold-biased mean temperatures that resemble climates from 15 to 20 years prior to the prediction period [Forecasting the Future with Yesterday's Climate: Temperature Bias in AI Weather and Climate Models]. This regression toward historical training data is not merely a static offset but is most pronounced for the hottest predicted temperatures, suggesting that these models have limited exposure to modern extreme heat events within their training datasets. Similarly, evaluations of GraphCast and Pangu-Weather during 60 distinct heat waves across the contiguous United States reveal consistent cold biases on regional scales at lead times of 5 to 10 days before heat wave onset, although GraphCast generally demonstrates higher skill than both Pangu-Weather and traditional numerical ensemble systems in these scenarios [18].

The struggle of AI models to accurately forecast record-breaking extremes is further compounded when compared to traditional Numerical Weather Prediction (NWP) systems. In direct comparisons involving record-breaking heat, cold, and wind events, the European Centre for Medium-Range Weather Forecasts' High RESolution forecast (HRES) consistently outperforms leading AI models such as GraphCast, Pangu-Weather, and Fuxi [5]. AI models tend to underestimate both the frequency and intensity of these record-breaking events, exhibiting growing errors as the magnitude of the record exceedance increases. This underprediction is partly attributed to the reliance on symmetric loss functions, such as Mean Squared Error (MSE), which inherently lead to biased predictions and the smoothing of extreme values [102]. Consequently, deterministic AI models often regress toward climatology beyond a lead time of approximately 7 to 10 days, retaining synoptic-scale dynamical structure while losing the surface-impact amplitude necessary for accurate extreme event forecasting [13].

Despite these limitations, significant progress has been made in mitigating biases and enhancing predictive skill through specialized training strategies and architectural innovations. Studies demonstrate that training neural weather models with custom loss functions tailored to emphasize extremes, such as exponential losses, leads to significant improvements in heat wave prediction without compromising general temperature forecasting accuracy [6]. Furthermore, the implementation of asymmetric optimization techniques and training-free enhancement modules like ExBooster has shown promise in increasing the hit rate of low-probability extreme events by capturing prediction uncertainty [102]. The adoption of generative approaches, specifically diffusion models, offers another pathway to overcome the "blurriness" of deterministic forecasts. Models like GenCast and FuXi-Extreme utilize diffusion processes to restore fine-scale details and generate realistic ensemble members, thereby better representing the tails of temperature distributions and improving the prediction of extreme weather compared to operational ensemble systems [163; 17].

The capacity of AI models to generalize to unseen climate states, often referred to as "gray swan" events, presents a nuanced picture. While some models struggle to extrapolate from weaker events in their training data to stronger, unseen extremes, others exhibit "translocation" capabilities, leveraging dynamically similar events from different regions to forecast unprecedented local phenomena [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall]. However, the absence of specific extreme categories, such as Category 3-5 tropical cyclones, in training data can severely degrade a model's ability to forecast those specific extremes, indicating that current learning strategies may not fully support robust extrapolation [25]. To address data scarcity for rare events, researchers have turned to massive ensemble simulations and long-term climate model outputs. For instance, generating huge ensembles using machine learning models has allowed for the simulation of extreme heatwave scenarios that exceed the prediction envelopes of smaller numerical ensembles, providing valuable storyline simulations for humid heat extremes [66]. Additionally, training on thousands of years of climate model data enables neural networks to identify precursors to long-lasting heatwaves, such as specific geopotential height patterns, even in regimes of historical data lack [86; 7].

Hybrid approaches and post-processing techniques further bridge the gap between AI efficiency and physical realism. Probabilistic bias correction frameworks have been shown to double the subseasonal skill of AI forecasting systems, effectively reducing systematic errors that plague long-range extreme event predictions [213]. Moreover, integrating physical constraints, such as terrain-following coordinates and conservation schemes, into AI architectures helps reduce drizzle bias and improves the estimation of extreme precipitation and temperature intensities [INVESTIGATING THE CONTRIBUTION OF TERRAIN-FOLLOWING COORDINATES AND CONSERVATION SCHEMES IN AI-DRIVEN PRECIPITATION FORECASTS]. While AI models currently face challenges in fully replicating the intensity of land-atmosphere feedbacks during extreme heatwaves due to the lack of interactive land components in some architectures, they offer unparalleled computational efficiency for generating large ensembles necessary for risk assessment and attribution studies [Testing NeuralGCM's capability to simulate future heatwaves based on the 2021 Pacific Northwest heatwave event; 67]. The ongoing development of interpretable models and the refinement of loss functions continue to enhance the reliability of AI-driven heatwave forecasts, positioning them as complementary tools to traditional NWP in the pursuit of accurate extreme weather prediction.

Precipitation and Convective Storms

The forecasting of extreme precipitation and convective wind gusts represents a critical frontier in meteorological science, where traditional Numerical Weather Prediction (NWP) models often struggle with computational costs and the resolution of fine-scale processes. Recent literature indicates that while global Artificial Intelligence (AI) models offer significant speed advantages, they frequently suffer from smoothing effects that underestimate the intensity of extremes. Consequently, a hybrid approach combining AI base models with sophisticated statistical and deep learning post-processing has emerged as a superior strategy for capturing spatial patterns and probabilistic distributions. For convective wind gusts, direct forecasting by Neural Weather Models (NWMs) is often outperformed by methods that empirically post-process predicted atmospheric environments. Research demonstrates that applying a hierarchy of statistical and deep learning techniques, particularly convolutional neural networks (CNNs) constrained by generalized extreme-value distributions, yields the most accurate hourly wind gust forecasts up to three days ahead [8]. This added value is crucial for designing responsive early-warning systems, as the post-processing step effectively translates coarse global environmental predictions into localized severe weather probabilities.

A primary limitation of standalone AI models is their tendency to produce "blurry" forecasts that overestimate drizzle while underestimating extreme precipitation intensities due to the smoothing inherent in training on reanalysis data with limited spatial resolution. To address this, generative adversarial networks (GANs) and diffusion models have been integrated into the forecasting pipeline to restore fine-scale structures. For instance, modifications to state-of-the-art models like FourCastNet using novel GAN architectures have achieved superior performance in capturing the extreme percentiles of global precipitation, matching NWP skill at short lead times [46]. Similarly, the FuXi-Extreme model employs a denoising diffusion probabilistic model to recover finer details in surface forecasts, significantly improving predictions of extreme total precipitation and wind speeds compared to both the base AI model and high-resolution NWP systems [17]. These generative approaches are essential because deterministic AI models often regress toward climatology at longer lead times, failing to maintain the amplitude of extreme events even when large-scale circulation patterns are correctly predicted [13].

The challenge of extrapolating to unseen extremes, or "gray swan" events, further underscores the necessity of robust post-processing and hybrid modeling. Studies reveal that AI models trained without exposure to major tropical cyclones struggle to extrapolate intensity for unprecedented events, suggesting a reliance on translocation of learned patterns rather than true physical extrapolation [25]. In such scenarios, post-processing frameworks that leverage physics-based NWP outputs as high-resolution data generators for training deep learning models prove vital. By fine-tuning deep learning models on different resolution outputs from physics-based simulations, researchers can better account for fine-scale processes that global datasets miss, thereby improving the prediction of tropical cyclone intensity and structure [231]. Furthermore, tracking-independent post-processing algorithms applied to NWM outputs have been shown to produce reliable tropical cyclone intensity forecasts up to five days ahead, extracting predictive information beyond what is encoded in initial conditions [3].

For convective storms specifically, the integration of ensemble methods with deep learning offers a pathway to quantify uncertainty and improve probabilistic guidance. Traditional ensemble prediction systems are computationally expensive, but mixed models utilizing deep neural networks to post-process a subset of trajectories can achieve comparable or superior skill with reduced costs [52]. Innovative approaches using conditional generative adversarial networks (CGANs) to create synthetic ensemble members from deterministic convection-allowing model forecasts have demonstrated skill increases of 10-20% in predicting severe weather hazards such as tornadoes and hail [284]. Additionally, transformer-based post-processing of AI forecasts has shown significant enhancements over dense neural networks by treating forecast lead times as sequential tokens, allowing the model to learn complex temporal relationships within the evolving atmospheric state [35].

Despite these advancements, contradictions and limitations persist regarding the reliability of AI models for record-breaking extremes. Comparative case studies indicate that while AI models may outperform NWP in predicting mean sea level pressure and general wind fields, they often underestimate peak wind amplitudes and fail to capture sharp frontal gradients associated with intense storms [60]. Moreover, for record-breaking heat, cold, and wind events, high-resolution NWP systems consistently outperform state-of-the-art AI models, highlighting current difficulties in extrapolating beyond the training domain [5]. Consequently, the most effective current paradigm involves hybrid

workflows where NWP outputs are fed into deep neural networks for bias correction and refinement, leveraging the physical consistency of dynamical models and the pattern-recognition efficiency of AI [79]. This synthesis is particularly evident in nowcasting, where physics-conditioned deep generative networks and attention-augmented convolutions outperform purely data-driven or purely physical baselines by preserving intense convective structures and mitigating rapid intensity decay [59] [21].

The Challenge of Extrapolation and "Gray Swan" Events

A fundamental limitation inherent to current data-driven artificial intelligence (AI) weather models is their struggle to reliably extrapolate to "gray swan" events-plausible but unobserved extremes that lie outside the training distribution. This inability to generalize from weaker, observed events to stronger, unseen extremes poses a critical risk for climate change adaptation, where future weather patterns may exceed historical precedents encoded in existing datasets. Empirical evidence suggests that AI models trained without specific categories of severe weather fail to forecast them accurately when they occur. For instance, studies utilizing the FourCastNet model demonstrate that versions trained on datasets with Category 3-5 tropical cyclones removed cannot accurately forecast Category 5 storms, indicating a failure to extrapolate intensity from weaker storm data [25]. This limitation is not unique to a single architecture; comparative analyses reveal that for record-breaking weather extremes, numerical models like the European Centre for Medium-Range Weather Forecasts' High RESolution forecast (HRES) consistently outperform state-of-the-art AI models such as GraphCast, Pangu-Weather, and Fuxi [5]. These AI systems tend to underestimate both the frequency and intensity of record-breaking events, often exhibiting larger forecast errors for heat, cold, and wind extremes as the magnitude of the record exceedance grows.

The mechanism behind these failures often involves a regression toward climatology, particularly at longer lead times. While models like Aurora demonstrate strong short-range skill, a consistent failure mode emerges in the subseasonal range where threshold-based extreme intensity collapses even as large-scale circulation patterns remain skillful [13]. This phenomenon is partly attributed to the optimization strategies employed during training. The widespread use of symmetric loss functions, such as Mean Squared Error (MSE), biases predictions toward the mean, leading to smoothed forecasts that systematically underestimate extreme values [102]. Furthermore, spectral bias in the optimization process can prevent models from learning the high-frequency components necessary to resolve sharp gradients associated with extremes, a problem distinct from simple data imbalance [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall].

However, the landscape of AI extrapolation is nuanced, with some models demonstrating success through "translocation" rather than pure extrapolation. In the case of the unprecedented 2024 Dubai rainfall, GraphCast successfully forecasted the event by leveraging dynamically similar events from other regions present in its global training data, effectively translocating knowledge rather than extrapolating from weaker local events [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall]. This suggests that while pure extrapolation from weaker to stronger intensities within a single regime remains a significant hurdle, global models may possess an implicit ability to generalize across basins if analogous dynamical structures exist elsewhere in the training set. Despite this, the general tendency for AI models to drift toward their training climatology remains a persistent issue. When evaluated under future climate scenarios, several models exhibit a cold bias, drifting toward the present-day climate conditions they were trained on, which compromises their utility for projecting future extremes in a warming world [245] [Forecasting the Future with Yesterday's Climate: Temperature Bias in AI Weather and Climate Models].

The challenge is further compounded by the resolution limits of training data. Global AI models are often trained on reanalysis data with coarse spatial resolution, which smooths out key features of weather systems such as the eye walls of tropical cyclones. Consequently, these models struggle to resolve the mesoscale structures critical for predicting intensity, often requiring post-processing or hybrid approaches to recover fine-scale details [3] [231]. Even when models capture the synoptic structure of a storm, they frequently underestimate peak amplitudes, such as wind speeds in extratropical cyclones, and fail to resolve sharp frontal gradients essential for impact warnings [60]. To address these deficits, researchers are exploring novel loss functions that asymmetrically optimize for extremes, diffusion models to restore fine-scale details, and hybrid frameworks that integrate physics-based constraints to prevent unphysical drift [17] [31]. Nevertheless, the consensus remains that without novel learning strategies or hybridization with physical laws, purely data-driven models face significant barriers in providing reliable early warnings for the rarest and most impactful weather events that define the gray swan category.

Out-of-Distribution Generalization Capabilities

The capacity of artificial intelligence (AI) weather models to generalize beyond their training distributions represents a critical frontier in meteorological science, particularly when addressing the "gray swan" events that define the upper tails of climate risk. A primary concern is whether these data-driven systems can extrapolate intensity scales from weaker, frequently observed events to stronger, unseen extremes, or if they are fundamentally limited to interpolating within the bounds of their training data. Research utilizing the FourCastNet model provides compelling evidence that while AI models possess robust spatial generalization capabilities across different tropical basins, they fail significantly when required to extrapolate intensity categories absent from the training set. Specifically, when FourCastNet was trained on historical data with Category 3-5 tropical cyclones removed, it retained the ability to forecast Category 5 storms in basins where such storms were present in the training data, suggesting that regional dynamical information is implicitly encoded and transferable [25]. However, the same model completely failed to forecast Category 5 intensity in basins where high-intensity storms were excluded from training, indicating an inability to extrapolate physical intensity scales from weaker analogs [25]. This limitation underscores a fundamental distinction between translocation-learning from dynamically similar events in other regions-and true extrapolation, a phenomenon further observed in the forecasting of Dubai's unprecedented 2024 rainfall, where GraphCast succeeded via translocation while FuXi, lacking comparable training examples, underestimated the event [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall].

The failure to extrapolate is not merely a function of data imbalance but is often rooted in the optimization objectives and architectural biases of current models. Many state-of-the-art AI weather prediction (AIWP) models exhibit a systematic low bias in tropical cyclone intensity, frequently reducing storm strength within the first 24 hours of the forecast, which renders them less accurate than simple climatology and persistence baselines for intensity prediction [243]. This tendency to regress toward the mean is exacerbated by the use of mean squared error loss functions, which favor spatial smoothing and blur extreme values, leading to "blurry" precipitation forecasts that underestimate drizzle and miss extreme intensities [42]. Consequently, for record-breaking extremes in heat, cold, and wind, numerical weather prediction (NWP) models like the ECMWF High RESolution forecast (HRES) consistently outperform AI counterparts such as GraphCast, Pangu-Weather, and FuXi, which struggle to forecast events that exceed the magnitude of anything seen during training [5]. This performance gap is particularly pronounced in subseasonal forecasting, where threshold-based extreme intensity collapses as fields regress toward climatology beyond a 7-to-10-day lead time, even if large-scale circulation patterns remain skillful [13].

Despite these limitations, several strategies have emerged to enhance out-of-distribution performance, primarily through hybrid approaches that integrate physical constraints or utilize generative architectures. The integration of physics-based NWP models as data generators for training deep learning systems allows for the representation of fine-scale processes critical for tropical cyclone intensity, which coarse global reanalysis datasets often smooth out [231]. Furthermore, hybrid frameworks that combine regional autoregressive prediction with large-scale dynamical constraints from global AI systems, such as the Hybrid Intelligent Typhoon System (HITS), have demonstrated substantial improvements in typhoon intensity forecasts by resolving mesoscale structures while maintaining dynamic consistency [109]. Generative models, including diffusion models and generative adversarial networks (GANs), offer another pathway by restoring fine-scale details and generating realistic extreme values that deterministic models typically suppress. For instance, FuXi-Extreme employs a denoising diffusion probabilistic model to restore surface forecast details, outperforming its deterministic baseline in extreme precipitation and wind forecasts, although it still lags behind HRES in tropical cyclone intensity [17]. Similarly, the Kunyu model integrates adversarial losses to reduce blurring and improve the estimation of anomaly extremes, challenging the dominance of regression-only losses [40].

The challenge of generalization extends to changing climate states, where models trained on present-day data must forecast future warmer climates. While some studies indicate that machine learning models can skillfully forecast across different climate states, suggesting that short-timescale dynamics may not differ fundamentally, others report a drift toward the training climatology, manifesting as a cold bias in future warmer scenarios [245]. This bias is evident in models like FourCastNet and Pangu-Weather, which show the strongest cold biases for the hottest predicted temperatures, reflecting limited exposure to modern extreme heat events during training [Forecasting the Future with Yesterday's Climate: Temperature Bias in AI Weather and Climate Models]. To mitigate these issues, researchers are

exploring concept-gated fine-tuning frameworks that selectively adapt models for failure cases without degrading performance on common scenarios, as well as physics-informed architectures that enforce conservation laws and terrain-following coordinates to better capture the dynamics of extreme events [31][42]. Ultimately, while AI models show promise in capturing synoptic-scale structures and translocating knowledge across regions, reliable forecasting of unprecedented extremes likely requires novel learning strategies that move beyond pure data-driven interpolation to incorporate explicit physical constraints and generative capabilities [25].

Generating Synthetic Extremes for Risk Assessment

The scarcity of observational data for high-impact, low-probability events, often termed "gray swans," presents a fundamental barrier to robust risk assessment and infrastructure design. To overcome the limitations of short historical records, researchers are increasingly turning to synthetic data generation methods that can extrapolate beyond observed conditions. A prominent approach involves the use of differentiable weather models to actively perturb initial conditions, thereby engineering plausible but unobserved extreme scenarios. For instance, the methodology described in [23] utilizes gradient descent on a loss function within a differentiable weather prediction model to alter the trajectory and intensity of ordinary cyclones. By minimizing this loss subject to constraints on perturbation size, the method successfully transformed Hurricane Fiona into a "gray swan" event resembling Hurricane Sandy, demonstrating that small perturbations to the extratropical state can trigger interactive strengthening and merger processes that yield significantly lower sea-level pressures than historically observed. This technique highlights the potential of dynamical models to explore the tails of hazard distributions that are absent from the observational record, although it is noted that such solutions do not inherently inform on the probability of occurrence.

Complementing these physics-based optimization techniques, deep learning approaches offer alternative pathways for generating synthetic storm databases. [36] details the creation of coupled trajectory and intensity models trained on the HURDAT2 database to simulate storm properties consistent with historical records. By advancing these models in six-hour increments, researchers can generate extensive synthetic databases from which frequencies of large wind speeds and other extreme properties can be extracted for specific domains. However, the ability of purely data-driven models to extrapolate to unseen extremes remains a critical area of investigation. Research presented in [25] indicates that current AI weather models, such as FourCastNet, struggle to extrapolate from weaker storms in their training set to predict Category 5 tropical cyclones if such events are excluded from the training data. This limitation suggests that while AI models can generalize across basins, they may lack the physical inductive biases necessary to reliably forecast extremes that lie outside their training distribution without novel learning strategies.

To address the tendency of AI models to produce smooth, blurred forecasts that underestimate extreme intensities, generative architectures such as diffusion models are being employed. [163] introduces a probabilistic model that generates sharp, realistic ensemble members by iteratively refining noise samples, avoiding the blurring effects common in mean-squared-error-trained models. Similarly, [17] utilizes a denoising diffusion probabilistic model to restore fine-scale details in surface forecasts, successfully improving the prediction of extreme precipitation and wind speeds compared to both the base FuXi model and high-resolution operational forecasts. These generative approaches are particularly valuable for risk assessment because they can produce diverse, physically plausible realizations of extreme events. Furthermore, [190] demonstrates how diffusion models like "Climate in a Bottle" can be guided to generate specific extreme states, such as tropical cyclones at user-defined locations, while simultaneously quantifying the likelihood of these guided samples relative to the unguided distribution. This capability allows for importance sampling of rare events, significantly reducing the computational cost of estimating extreme event probabilities compared to traditional Monte Carlo methods.

The integration of AI with physical models further enhances the generation of synthetic extremes by leveraging the strengths of both paradigms. Hybrid systems, such as the one described in [164], combine machine learning large-scale forecasts with physics-based mesoscale models using spectral nudging. This fusion improves the representation of typhoon intensity and structure, which is crucial for generating realistic synthetic scenarios for risk analysis. Additionally, [109] presents a hybrid framework that combines regional autoregressive prediction with large-scale dynamical constraints, substantially improving precipitation structure and intensity forecasts compared to purely autoregressive models. The use of such hybrid systems ensures that synthetic extremes maintain physical consistency, a feature often lacking in purely data-driven approaches as noted in [123], where purely data-driven models were found to disrupt key atmospheric balances during extreme events.

Despite these advancements, challenges remain in ensuring the robustness and physical fidelity of generated extremes. Studies such as [93] reveal that while AI models can preserve general storm trajectories under noisy initial conditions, they often systematically underestimate storm intensity. Moreover, [Watch an AI Weather Model Learn (and Unlearn) Tropical Cyclones] provides evidence that AI models may "unlearn" relationships between moisture and tropical cyclone intensity during later stages of training, potentially compromising their ability to simulate extreme

intensification. Consequently, while methods like gradient descent on differentiable models and generative diffusion frameworks offer powerful tools for populating the "gray swan" space, their application in risk assessment requires careful validation against physical principles and an understanding of their inherent biases regarding event probability and intensity. The development of frameworks like [19] underscores the need for interchangeable components that can rigorously test and refine these synthetic generation techniques to ensure they provide reliable inputs for emergency planning and infrastructure resilience.

Advanced Architectures and Post-Processing Techniques

The inherent smoothing effect observed in global Neural Weather Models (NWMs), primarily due to training on reanalysis datasets with coarse spatial resolutions such as 0.25° , necessitates the development of hybrid approaches that integrate specialized post-processing layers and advanced architectural designs. While models like Pangu-Weather and FourCastNet have demonstrated competency in medium-range forecasting, they frequently struggle to resolve fine-scale extrema critical for impact assessment, such as tropical cyclone intensity and convective wind gusts [3]. To mitigate these resolution limitations, researchers are increasingly adopting post-processing frameworks that operate on NWM outputs to recover lost high-frequency details. For instance, tracking-independent post-processing models applied to Pangu-Weather and FourCastNet v2 have successfully produced reliable probabilistic forecasts of tropical cyclone intensity up to five days ahead, extracting predictive information that linear models and simple multilayer perceptrons can access without the need for complex convolutional architectures [3]. Similarly, for convective storm wind gusts, applying a hierarchy of statistical and deep learning methods to the atmospheric environments predicted by NWMs has yielded superior results compared to direct forecasting, particularly when constraining probabilistic forecasts using generalized extreme-value distributions [8].

Beyond simple statistical correction, the field is witnessing a shift toward generative architectures capable of downsampling coarse global forecasts into high-resolution regional fields. Diffusion models and Generative Adversarial Networks (GANs) are emerging as powerful tools for this purpose. The FuXi-Extreme model, for example, employs a denoising diffusion probabilistic model to restore finer-scale details in surface forecast data, effectively addressing the underestimation of extreme precipitation and wind speeds common in standard ML forecasts [17]. In the context of typhoons, the Intelligent Shanghai Typhoon Model (ISTM) utilizes a two-stage UNet-Diffusion framework to downscale coarse ERA5 reanalysis to kilometer-scale resolution, significantly outperforming baseline regression models in capturing storm structure and intensity [132]. Furthermore, generative approaches are being used to create synthetic ensemble members from deterministic convection-allowing model outputs, enhancing the probabilistic prediction of severe weather hazards like tornadoes and hail [284]. These generative techniques offer a pathway to produce "infinite ensembles" and high-resolution fields that traditional numerical weather prediction (NWP) systems cannot generate at comparable computational costs [151].

Architectural innovations are also focusing on integrating physical constraints directly into the neural network design to improve consistency and reduce artifacts. Hybrid models that decompose weather forecasting into distinct advection and convection components have shown promise in simulating atmospheric transitions more efficiently than unified models [162]. Additionally, incorporating physics-informed inductive biases, such as terrain-following coordinates and conservation schemes, has been shown to reduce drizzle bias and improve the estimation of extreme precipitation events [INVESTIGATING THE CONTRIBUTION OF TERRAIN-FOLLOWING COORDINATES AND CONSERVATION SCHEMES IN AI-DRIVEN PRECIPITATION FORECASTS]. The YingLong model exemplifies this trend by employing a parallel structure of global and local blocks to capture multiscale features in limited-area forecasting, demonstrating superior skill in surface wind speed prediction compared to dynamical models like WRF-ARW [68]. Despite these advancements, challenges remain regarding the physical consistency of data-driven models. Studies indicate that while regional deep learning models may achieve low error metrics, they can disrupt key atmospheric balances and fail to capture mesoscale features during extreme events, raising concerns about their operational reliability without further physical constraints [123].

The integration of NWP outputs with observational data represents another critical avenue for improving local forecast accuracy. Hybrid neural networks that combine NWP predictions with recent past station observations have demonstrated significant reductions in root mean square error for short-term wind speed forecasting, particularly in complex terrains [14]. This approach leverages the strengths of both physics-based simulations and real-time data to correct biases and capture localized variations. Furthermore, transformer-based post-processing techniques are being explored to treat forecast lead times as sequential tokens, allowing the model to learn complex temporal relationships within the evolving atmospheric state and significantly enhancing medium-range severe weather guidance [35]. However, the efficacy of these advanced architectures is contingent upon the quality of training data and the ability to generalize to out-of-distribution extremes. Research suggests that current AI models often struggle to extrapolate to record-breaking events or "gray swan" tropical cyclones that were absent from their training datasets, highlighting the

need for novel learning strategies and robust verification frameworks like the Extreme Weather Bench to ensure trustworthiness [25; 2; 5]. Ultimately, the future of high-impact weather forecasting lies in the seamless integration of data-driven efficiency with physical realism, utilizing hybrid systems that can dynamically adapt to both global contexts and local extremes.

Diffusion Models and Generative Approaches

The integration of denoising diffusion probabilistic models (DDPM) and other generative architectures into weather forecasting represents a paradigm shift aimed at resolving the systematic underestimation of extreme weather events inherent in deterministic artificial intelligence (AI) models. Traditional machine learning approaches, often optimized using symmetric loss functions like Mean Squared Error (MSE), tend to produce overly smooth predictions that regress toward the climatological mean, thereby failing to capture the high-frequency variability and intensity of phenomena such as tropical cyclones, heatwaves, and heavy precipitation. Generative approaches address this "blurring" effect by learning the underlying probability distribution of atmospheric states, allowing for the synthesis of realistic, high-resolution ensemble members that preserve physical extremes. A primary application of this technology is the post-processing of coarse global forecasts to restore fine-scale details. For instance, the FuXi-Extreme model employs a DDPM to correct surface forecasts generated by the deterministic FuXi model, effectively restoring finer-scale details in 5-day forecasts and significantly improving the prediction of extreme total precipitation and wind speeds compared to both the base AI model and high-resolution numerical weather prediction (NWP) systems [17]. Similarly, the Intelligent Shanghai Typhoon Model (ISTM) utilizes a two-stage UNet-Diffusion framework to downscale coarse reanalysis data to kilometer-scale resolution, successfully capturing the structure and intensity of typhoons where standard AI models typically fail [132].

Beyond post-processing, diffusion models are being deployed as core components of end-to-end ensemble forecasting systems, offering a computationally efficient alternative to traditional perturbation-based ensembles. GenCast, a diffusion-based ensemble forecasting model, generates global 15-day forecasts that demonstrate superior skill over operational NWP ensembles, particularly in predicting extreme weather and tropical cyclone tracks. By iteratively refining noise samples conditioned on previous atmospheric states, GenCast produces sharp, realistic samples that avoid the spectral energy loss characteristic of deterministic models, ensuring that nonlinear derived quantities like wind speed remain unbiased [163]. This capability extends to latent space operations, as seen in LaDCast, which operates entirely within a learned latent space to generate hourly ensemble forecasts. LaDCast achieves probabilistic skill comparable to major operational centers while drastically reducing computational costs, demonstrating superior performance in tracking rare extreme events such as cyclones [49]. Furthermore, the Anemoui framework implements a probabilistic diffusion-based method for global atmospheric downscaling, transforming low-resolution ensemble forecasts into high-resolution ensembles by learning the conditional distribution of finer-scale residuals, thereby reproducing target power spectra and capturing physically consistent multivariate relationships [30].

A distinct advantage of diffusion models lies in their ability to estimate probability densities, enabling rigorous quantification of extreme event likelihoods and facilitating targeted scenario generation. The Climate in a Bottle (cBottle) emulator illustrates how diffusion models can be guided to generate specific extreme states, such as tropical cyclones in user-defined locations, while simultaneously calculating the likelihood of these events through probability flow ordinary differential equations. This allows for importance sampling that reduces the standard error of probability estimates for rare events, a capability crucial for climate adaptation and risk assessment [190]. In the realm of subseasonal-to-seasonal (S2S) forecasting, the TianXing-S2S model incorporates a diffusion module within a multi-sphere coupled framework to generate daily ensemble forecasts, outperforming existing systems in predicting heat waves and anomalous precipitation by effectively modeling intrinsic atmospheric uncertainty [269].

Despite these advancements, challenges remain regarding the physical consistency and extrapolation capabilities of generative models. While diffusion models excel at generating realistic textures and extremes within the training distribution, concerns persist about their ability to forecast unprecedented "gray swan" events or maintain physical balances over long rollouts. Studies indicate that while some models like GraphCast can generalize across basins via translocation, others struggle to extrapolate beyond training data magnitudes, often underestimating record-breaking extremes [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall] [5]. Additionally, purely data-driven generative models may occasionally violate conservation laws or disrupt atmospheric balances, as observed in regional stretched-grid models that produce fine-scale noise inconsistent with physical dynamics [123]. To mitigate these issues, hybrid approaches combining diffusion models with physics-guided constraints or using them as emulators for physics-based systems are gaining traction. For example, Nowcast3D integrates physically constrained neural operators with a conditional diffusion model

to ensure physical plausibility in 3D precipitation nowcasting, while other frameworks employ adversarial losses alongside regression objectives to enhance forecast realism and reduce blurring [178] [40]. The development of specialized loss functions, such as Exloss, further complements generative approaches by explicitly optimizing for extreme values during training, addressing the spectral bias that often plagues symmetric optimization strategies [102]. Collectively, these developments suggest that while diffusion models offer a powerful mechanism for restoring extreme variability, their operational reliability depends on careful integration with physical priors and robust uncertainty quantification frameworks.

Hybrid and Nested Modeling Frameworks

The evolution of deep learning in meteorology has increasingly shifted toward hybrid and nested modeling frameworks designed to reconcile the computational efficiency of data-driven approaches with the physical fidelity required for high-resolution regional forecasting. A primary challenge in this domain is balancing global context with local detail, particularly when predicting extreme events that manifest at fine spatial scales. OneForecast addresses this by proposing a global-regional nested framework based on graph neural networks (GNNs), which constructs a multi-scale graph structure to densify target regions and capture local high-frequency features while maintaining global coherence [12]. This approach utilizes an adaptive messaging mechanism with dynamic gating units to integrate node and edge features, effectively mitigating the excessive smoothing often observed in extreme event predictions. Similarly, the concept of nested grids is explored in regional data-driven modeling, where a stretched-grid architecture dedicates higher resolution to specific areas of interest, such as the Nordics, while maintaining coarser resolution elsewhere [287]. These models leverage the inherent flexibility of GNNs to handle arbitrary multi-resolution configurations, demonstrating that pre-training on global data followed by refinement on high-resolution operational analyses can outperform traditional ensemble means for surface variables.

Beyond static nesting, hierarchical representations have emerged as a critical strategy for preserving physical consistency across scales. Hierarchical Graph Networks, exemplified by HiFlowCast and HiAntFlow, introduce a Latent-Memory-Retention mechanism to prevent the erosion of global trends during downward traversal in multiscale frameworks [142]. By embedding physics within the prediction process through a Latent-to-Physics branch, these models ensure that nonlinear mappings do not disconnect regional forecasts from large-scale drivers, resulting in significant error reductions for extreme quantiles. Complementing this, STCast introduces an adaptive boundary alignment mechanism to overcome the limitations of static regional boundaries often found in cropping-based methods [280]. Through a Spatial-Aligned Attention mechanism, STCast dynamically refines boundaries based on attention-derived patterns, allowing for more robust generalization across different atmospheric regimes. These advancements highlight a consensus in the literature that rigid separation between global and regional models leads to information loss at boundaries, necessitating adaptive or hierarchical integration strategies.

The integration of physics-based numerical weather prediction (NWP) outputs into deep learning frameworks represents another vital dimension of hybrid modeling. Rather than viewing NWP and AI as competing paradigms, recent studies propose using NWP models as high-fidelity data generators to train deep learning systems for regional extremes [231]. This is particularly relevant for phenomena like tropical cyclones, where global AI models often lack the resolution to resolve fine-scale intensity structures. Hybrid operational models, such as FuXi-SHTM, combine the large-scale forecasting robustness of machine learning models with the mesoscale strengths of physics-based typhoon models through spectral nudging [164]. This dual physics-data-driven approach enhances track and intensity predictions for super typhoons, suggesting that the future of operational forecasting lies in tightly coupled systems rather than purely data-driven replacements. Furthermore, post-processing techniques serve as a lightweight hybrid layer, where neural networks correct biases in ensemble forecasts or downscale low-resolution outputs to capture extreme values more accurately [180; 30].

Despite these advances, contradictions and limitations persist regarding the ability of hybrid models to extrapolate beyond their training distributions. While some studies indicate that global effective receptive fields allow models like GraphCast to successfully forecast "gray swan" events through translocation of learned dynamics from other regions [Predicting Beyond Training Data via Extrapolation versus Translocation: AI Weather Models and Dubai's Unprecedented 2024 Rainfall], others argue that numerical models still consistently outperform AI in predicting record-breaking extremes due to the latter's tendency to regress toward climatology [5; 10]. The discrepancy often stems from the smoothing effect inherent in many deep learning architectures, which can underestimate the amplitude of extreme winds and temperatures even when the synoptic structure is correctly captured [60]. To counteract this, specialized loss functions and generative components, such as diffusion models in FuXi-Extreme, are being integrated to restore fine-scale details and improve the representation of tail distributions [17; 102]. Additionally, the lack of interactive land components in many atmospheric-only AI models limits their ability to simulate feedback loops crucial for heatwave intensification, a gap that hybrid storyline analyses attempt to address by constraining AI dynamics with

physical nudging [Testing NeuralGCM's capability to simulate future heatwaves based on the 2021 Pacific Northwest heatwave event].

The development of these frameworks also emphasizes the need for rigorous verification standards that account for spatial coherence and decision thresholds. Traditional global metrics often fail to capture the localized skill of nested models, prompting the creation of threshold-weighted spatial verification approaches that better evaluate extreme event performance [130]. As the field moves toward universal foundation models, the integration of adaptive frequency modulation and event prior augmentation becomes essential to handle the spectral disparity between normal and extreme weather regimes [26]. Ultimately, the synthesis of graph-based nesting, hierarchical memory retention, and physics-informed post-processing constitutes the current frontier in achieving reliable, high-resolution forecasts for high-impact weather events.

Custom Loss Functions and Transfer Learning

Within the broader domain of advanced architectures and post-processing techniques, the optimization of training objectives has emerged as a critical lever for enhancing the fidelity of data-driven weather forecasts, particularly regarding extreme events. A pervasive limitation in standard machine learning weather prediction (MLWP) models is the tendency toward spatial and temporal smoothing, often referred to as "blurring," which results in the systematic underestimation of extreme values. This phenomenon is mathematically rooted in the widespread reliance on symmetric loss functions, such as Mean Squared Error (MSE), which penalize over- and under-predictions equally, thereby biasing forecasts toward the conditional mean of the distribution [102]. Research indicates that minimizing MSE leads to forecasts that lose high-frequency power and fail to capture the intensity of record-breaking heat, cold, and wind events, creating a significant gap between AI models and numerical weather prediction (NWP) systems in high-stakes scenarios [5].

To mitigate this smoothing effect, recent studies have successfully implemented custom, asymmetric loss functions designed to explicitly prioritize extreme values. For instance, the introduction of "Exloss," an asymmetric optimization function, has been shown to highlight extreme values during training, effectively correcting the bias inherent in symmetric losses and improving the prediction of extreme weather without compromising overall accuracy [102]. Similarly, in the context of global heat wave forecasting, neural weather models trained with exponential losses tailored to emphasize extremes have demonstrated significant skill improvements compared to those trained on MSE. Notably, the use of a symmetric exponential loss was found to reduce the smoothing of forecasts as lead time increases, allowing models to maintain sharper temperature anomaly predictions over medium to subseasonal ranges [6]. These findings suggest that the choice of loss function is not merely a hyperparameter tuning exercise but a fundamental architectural decision that dictates a model's ability to resolve the tails of the atmospheric distribution.

Complementing the modification of loss functions, transfer learning has proven to be an efficient strategy for adapting general-purpose weather models to specific extreme event tasks. Rather than training models from scratch, which is computationally prohibitive, researchers have demonstrated that re-training existing neural weather models with custom loss functions for only a few epochs can yield substantial gains in extreme event prediction. This approach allows models to retain their robust general temperature forecasting capabilities while acquiring specialized sensitivity to heat waves [6]. The efficacy of transfer learning extends to addressing class imbalance issues inherent in rare event prediction; by leveraging pre-trained weights and applying techniques such as large-class undersampling, convolutional neural networks can effectively forecast long-lasting extreme heatwaves up to 15 days in advance [7]. Furthermore, transfer learning facilitates the adaptation of models to changing climate conditions, although challenges remain regarding the generalization of models trained on historical data to future, warmer climate states where cold biases may persist [Forecasting the Future with Yesterday's Climate: Temperature Bias in AI Weather and Climate Models].

Despite these advancements, contradictions and limitations persist within the literature. While some studies report that custom losses allow AI models to compete with or even surpass state-of-the-art physics-based systems in specific extreme metrics [6], others caution that AI models still consistently underperform NWP systems like ECMWF's HRES when forecasting unprecedented, record-breaking extremes that lie outside the training distribution [5]. This discrepancy highlights a tension between improving statistical skill on known extreme regimes and achieving true extrapolation capability for novel events. Additionally, the integration of adversarial losses alongside regression losses, as seen in the Kunyu model, offers another pathway to enhance forecast clarity and realism, suggesting that a hybrid loss framework may be superior to relying solely on custom regression targets [40].

The interplay between loss function design and model architecture further complicates the landscape. For example, while diffusion-based models like GenCast and FuXi-Extreme utilize generative objectives to naturally avoid the blurring associated with MSE, they represent a paradigm shift away from deterministic regression entirely [163][17]. In contrast, deterministic models relying on custom losses must carefully balance the weighting of extreme samples to prevent instability or degradation in general forecast skill. The success of transfer learning in this context underscores the importance of initializing models with robust dynamical representations before fine-tuning for extremal behavior. Ultimately, the synthesis of asymmetric loss functions, transfer learning protocols, and generative approaches represents a frontier in closing the utility gap between data-driven and physics-based forecasting, ensuring that AI models can reliably inform decision-making for increasingly frequent and severe weather hazards.

Conclusion

The integration of deep learning into meteorology has undeniably initiated a transformative era in weather forecasting, characterized by unprecedented computational efficiency and competitive skill in medium-range global predictions. However, a critical dichotomy remains between the performance of these data-driven models on aggregate metrics and their reliability in forecasting high-impact extreme events. While artificial intelligence systems excel at capturing synoptic-scale structures and ordinary atmospheric trajectories, they frequently struggle with the non-linear dynamics, sharp gradients, and rare intensities that define catastrophic weather. This limitation is largely attributed to the prevalence of symmetric loss functions and the reliance on historical reanalysis data, which induce a regression toward climatology and result in "smooth" forecasts that systematically underestimate the magnitude of record-breaking heat, wind, and precipitation events. Consequently, purely data-driven approaches currently lag behind traditional numerical weather prediction systems when tasked with extrapolating beyond their training domains to predict unprecedented "gray swan" phenomena.

To bridge this gap between statistical efficiency and physical fidelity, the research community is increasingly converging on hybrid methodologies that synthesize the speed of machine learning with the robustness of physical laws. Advances in generative modeling, particularly through diffusion processes and adversarial networks, have shown significant promise in restoring fine-scale details and generating realistic ensemble spreads that deterministic models fail to capture. Simultaneously, the development of asymmetric loss functions, physics-informed architectures, and nested modeling frameworks is enabling models to better respect conservation laws and resolve local extrema. These methodological innovations are complemented by a crucial shift toward standardized, impact-centric benchmarking. The establishment of specialized datasets and evaluation suites focused on extreme events is essential to move beyond global average scores, ensuring that model improvements translate into tangible gains in societal resilience and early warning capabilities.

Ultimately, the future of operational weather forecasting lies not in the replacement of numerical models but in their seamless integration with artificial intelligence to create universal, probabilistic Earth system emulators. Achieving this vision requires resolving persistent challenges related to out-of-distribution generalization, uncertainty quantification, and the interpretability of black-box decisions. By prioritizing physical consistency, enhancing the representation of compound events, and rigorously validating models against high-stakes scenarios, the field can evolve from producing accurate average states to delivering reliable, actionable forecasts for the most dangerous weather phenomena. This synthesis of data-driven pattern recognition and dynamical rigor represents the necessary pathway to safeguarding communities against the escalating risks of a changing climate.